

Farm Power and Machinery

Last 5-year PYQ from 2019-2023

1. Which of the following can be used for the production of bioethanol?

- a. Cassava, Horse gram, Potato, Sugarcane
- b. Horse-gram & legumes
- c. Carrot, beans, potato and castor
- d. None of the above

2. All the equipment's used for primary tillage are called as a

- a. Harrow
- b. Black hoe dozer
- c. Plough
- d. All of the above

3. How much horse power can a bullock generates

- a. 0.5 hp to 10 hp
- b. 0.50 hp to 0.75 hp
- c. 10 hp to 15 hp
- d. 0.25 hp to 5 hp

4. Which of the following equipment is used to sow seeds manually by make a hole in the soil

- a. Plough
- b. Dibbler
- c. Seed drill
- d. All of the above

5. Which of the following equipment is used for sowing seeds in a zero-tillage system

- a. Seed cum ferti drill
- b. Zero seed till drill
- c. Dibbler
- d. Transplanter

In 2023 mains Passage on tillage was given out of which following question were asked you may refer to the primary tillage topic of these notes for the passage

- 1. The major preparatory operation performed on the field to open up the hard soil and breaking the soil clods, cutting weeds and tilling the soil to the depth is called as _____**
 - a. Secondary tillage
 - b. Primary tillage
 - c. Deep tillage
 - d. None of the above

- 2. Which of the following equipment are used for primary tillage of the soil?**
 - a. Plough, harrow, cultivator
 - b. Disc plough, rotatory plough and cultivator
 - c. Cultivator, and harrow
 - d. Plough, disc plough, mouldboard plough

- 3. Tillage operations performed on the field in a standing crop are called as**
 - a. Preparatory tillage
 - b. After cultivation tillage
 - c. Secondary tillage
 - d. Subsoiling

- 4. _____ Equipment is used for shallow cultivation operations such as preparation of seedbed, covering seeds, and destroy weed seedlings.**
 - a. Rotatory plough
 - b. Harrow
 - c. Chisel plough
 - d. Plank

In 2023 an essay in English descriptive paper on topic Farm mechanisation in India and constrains in adoption of farm mechanisation on large scale.

Answers

1. A 2.C 3. B 4.B 5.B

Answers of passage-based question

1.B 2.D 3.B 4.B

Farm Power

Farm Power is an **essential input in agriculture** for timely field operations **for increasing production and productivity of land**. Farm power is used for operating different types of machinery like **tillage, planting, plant protection, harvesting and threshing machinery and other stationary jobs like operating irrigation equipment, threshers/shellers / cleaners/ graders, etc.**

Definition And Formula of Force, Work, Power and Energy

Force: An invisible agent which always tries to **change the state of the body**. The unit for force in **SI system is Newton (N)** and in MKS system is Kg-f

Work: When a force is applied to the body, it moves in the direction of the applied force, then work is said to be done. **The unit for work is Newton- meters (N-m).**

$$\text{Work done} = \text{Force} \times \text{Distance travelled (N-m)}$$

Power: It is the **rate of doing work**. The work done per unit time is called power and is expressed in **N-m-Sec-1**.

$$\text{Power} = \frac{\text{Work done}}{\text{Time taken}}$$

Energy: It is the **capacity to do the work**. The unit for energy is **Joules or Calories** or Kilo Watt or Horse Power.

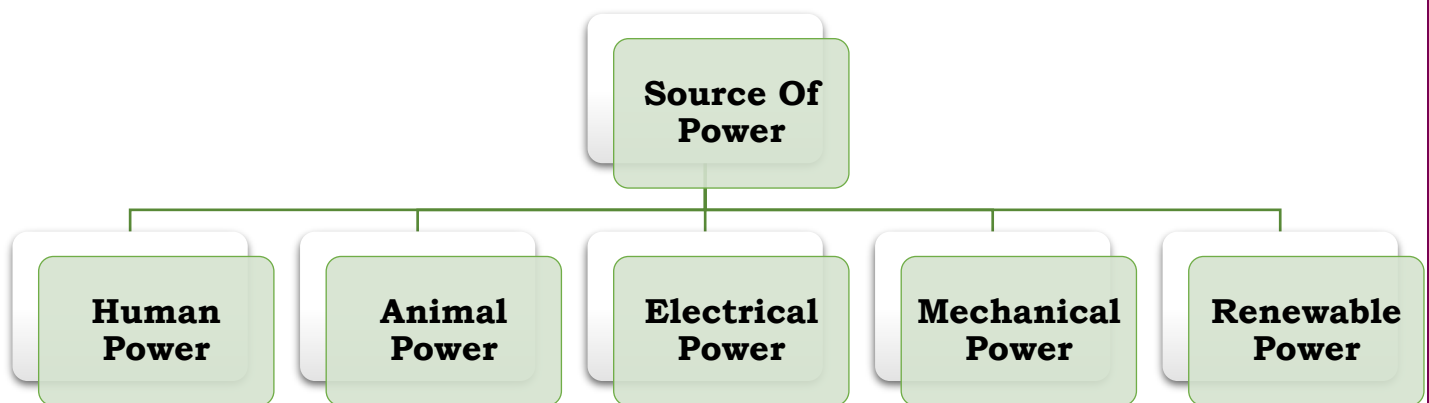
Horse Power: It is that amount of force which is **capable of displacing 75 kg force** through a distance of one meter in one second.

Sources Of Farm Power and Mechanization

Various types of agricultural operations performed on a farm can be broadly classified as:

- Tractive work such as seed bed preparation,
- Cultivation,
- Harvesting and transportation,
- Stationary work like silage cutting,
- Feed grinding,
- Threshing,
- Winnowing
- Lifting of irrigation water.

These operations are performed **by different sources of power** for doing these operations **different types of power available is classified as:**



1. Human Power

Human power on farm is in the **form of labour**

Labour (Human Energy) on Farms:

Labour is one of the most important **sources of farm power in regions where** traditional system of agriculture is practiced. **On small farms, high proportion of labour is supplied**

by the farmer and his family on small farm having very little spare capital to buy appropriate type of hand tools and animal drawn equipment, both labour use efficiency and productivity are very low.

Labour use efficiency can be improved by engaging labour in a group where sequence of operations demands teamwork for effective output.

For example, a power thresher operation always demands a **team effort** for efficient utilization of expensive resources i.e., **thresher, cleaner, the prime mover, etc.**



Figure 1 – Human labour

2. Animal Power

Animal power is the **most important source of power on the farm all over the world** particularly in developing countries. It is estimated that **nearly 80 per cent of the total draft power used in agriculture throughout the world is still provided by animals.**



Figure 2 – Animal power

Different animal sources are:

- **Bullocks-** can pull of **about 15% of its** weight
- **Buffaloes**
- **Camels**
- **Horses**
- **Donkeys-****can pull 80 %** of its weight for short period and 10-15% of its weight for sustainable period.
- **mules**
- **elephants**

The average force a bullock can exert is nearly equal to one tenth of its body weight. But for a very short period, it can exert many more times the average force. **Generally, a medium size bullock can develop between 0.50 to 0.75 hp.**

3. Mechanical Power

The **third important source of farm power is mechanical power** that is available through tractors and stationary engines. The engine is a highly efficient device for converting fuel into useful work.



Figure 3 – Mechanical Power

- **The efficiency of diesel engine varies between 32 and 38 per cent, whereas that of the petrol engine in the range of 25 and 32 per cent.**
- In recent years, **diesel engines and tractors have gained considerable popularity in Agri-cultural operations. Small pumping sets within 3 to 10 hp range** are very much in demand.
- Likewise, engines of **low to medium speed developing about 14 to 20 hp are successfully used for flourmills, oil expellers etc.**
- **Diesel engines of the larger size are used on tractors.** Diesel engines are the main source of power in agriculture.

4. Electrical Power

Now-a-days electricity has become a very important source of power on farms in various countries. It is steadily becoming more and more available with the increase of various river valley projects and thermal stations. **The largest use of electric power in the rural areas is for irrigation and domestic water supply.** Besides



Figure 4 – Electrical Power

this, **the use of electric power in dairy industry, cold storage, fruit processing and cattle feed grinding has tremendously increased.**

5. Renewable Energy – Wind Power, Solar Power Etc

A. Wind Power

The availability of **wind power for farm work is quite limited.** Where the wind velocity **is more than 32 km/h, wind mills can be used for lifting water.** The most important reason of its low use is its uncertainty. Thus, the average capacity of a **wind mill would be about 0.50 hp.** It is **one of the cheapest sources** of farm power available.



Figure 5 - Wind Mill

B. Solar Energy

Solar energy is the radiation from the **Sun capable of producing heat, causing chemical reactions, or generating electricity.**

The total amount of solar **energy received on Earth is vastly more than the world's current and anticipated energy requirements.** If suitably harnessed, solar energy has the potential to satisfy all future energy needs. In the **21st century solar energy has become increasingly attractive as a renewable energy source** because of its inexhaustible supply and its non-polluting character, in stark contrast to the finite fossil fuels coal, petroleum, and natural gas.



Figure 6 - Solar Irrigation Pump

Uses -Solar dryers, lantern, cooker, solar still, solar refrigeration, solar lighting, solar irrigation pumps, solar tubewells etc

Biomass Energy

Biomass is **renewable organic material that comes from plants and animals**. Biomass **contains stored chemical energy** from the sun that is produced by plants through photosynthesis. **Biomass can be burned directly for heat or converted** to liquid and gaseous fuels through various processes

Biomass Sources for Energy Include

- **Wood and wood processing waste**—firewood, wood pellets, and wood chips, lumber and furniture mill sawdust and waste, and black liquor from pulp and paper mills
- **Agricultural crops and waste materials**—**corn, soybeans, sugar cane**, switchgrass, woody plants, algae, and crop and food processing residues, mostly to produce biofuels
- **Biogenic materials in municipal solid waste**—paper products; cotton and wool products; and food, yard, and wood wastes
- **Animal manure and human sewage for producing biogas** (renewable natural gas)

Biofuel

Biofuel is a type of fuel that **is generated naturally or artificially from the biomass** of plant or algae material, and animal wastes. It is considered a renewable energy source because the feedstock material can be replenished easily. **They can be in solid, liquid, and gaseous forms but the liquid biofuels have the most potential** to replace conventional transportation fuels.

With rising crude oil prices, biofuels are seen as a sustainable alternative to be used mainly as a blending with petroleum-based fossil fuels. **India also has set targets of 20% ethanol blending in petrol and 5% biodiesel blending in diesel by 2030 under the National Policy on Biofuels 2018.**

Types of Biofuels

- 1. Biodiesel:** Biodiesel is a liquid fuel produced from vegetable oils and animal fats through transesterification produced from Jatropha plant seeds, castor seeds etc.
- 2. Bioethanol:** Bioethanol is ethyl alcohol (Ethanol) generated through microbial fermentation of plant materials like corn, sugarcane or agricultural residues using *Saccharomyces cerevisiae*.

- 3. Biogas:** Biogas is a **renewable fuel produced by anaerobic digestion of organic matter**, such as food or animal waste, in an oxygen-free environment, either naturally occurring or intentionally created for industrial use.
- 4. Bio CNG:** **Bio CNG is a purified form of biogas**, processed to produce 95% pure methane gas, similar to commercially available natural gas
- 5. Biochar:** It is the **spectrum of black carbon**, produced by the controlled process - pyrolysis of organic materials from forests and agricultural wastes.
- 6. Renewable Diesel:** **Previously known as green diesel**, chemically resembles petroleum diesel but is made from biomass feedstocks

Generations of Biofuels

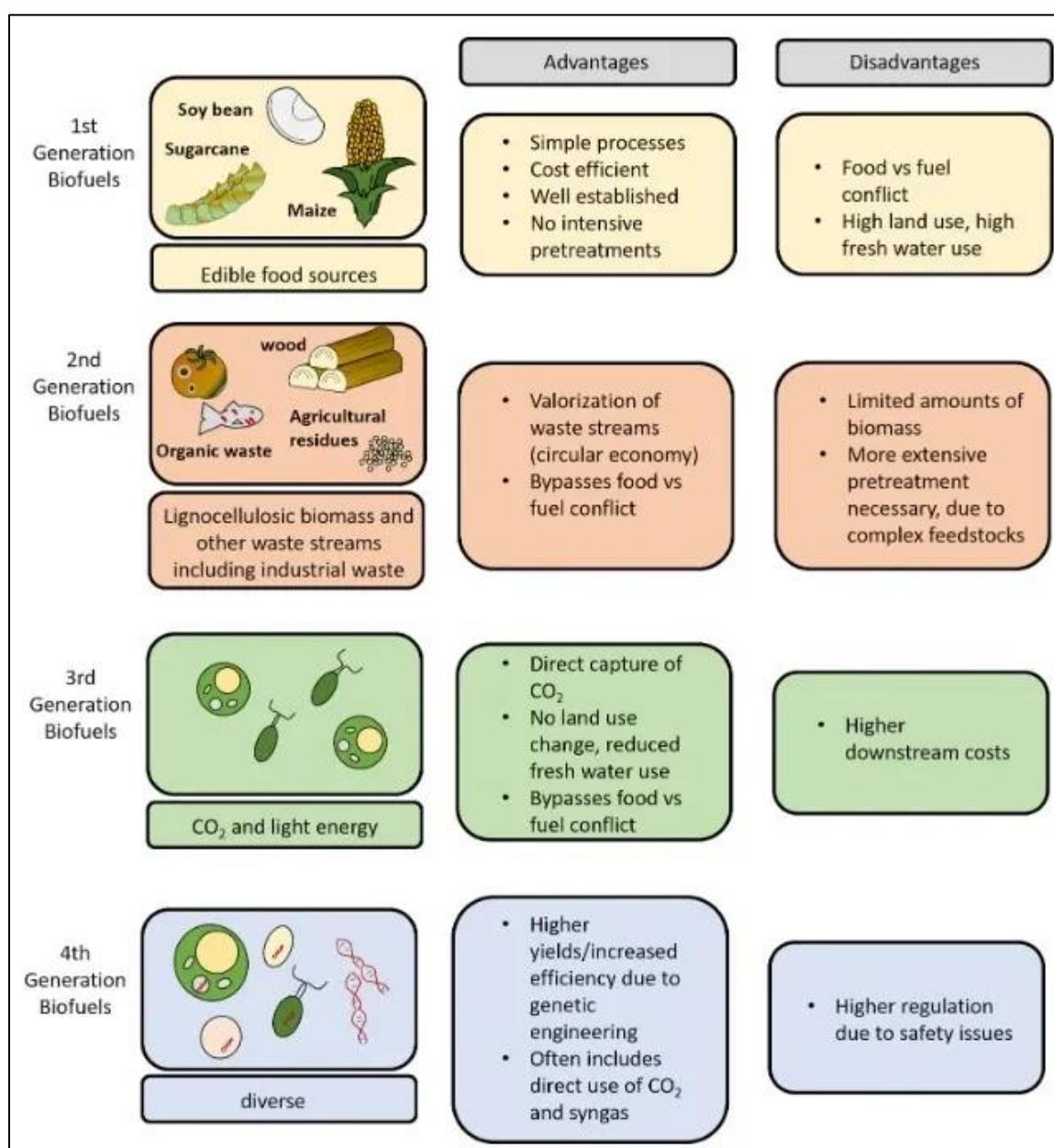


Figure 6
Biofuels Production

Farm Mechanization

Farm mechanization **is the application of engineering and technology in agricultural operations, to do a job in a better way to improve productivity.** This includes development application and management of **all mechanical aids for field production, water control, material handling, storing and processing.**

Mechanical aids include hand tools, animal drawn equipment, power tillers, tractors, engines, electric motors, processing and hauling equipment.

Scope Of Farm Mechanization

There is a good scope of farm mechanization in India due to the following factors:

- **Improved irrigation facility** in the country.
- Introduction of **high yielding varieties of seeds.**
- Introduction of **high dose of fertilizers and pesticides** for different crops.
- Introduction of **new crops in different parts of the country.**
- **Multiple cropping system and intensive cultivation** followed in different parts of the country.

Factors Responsible To Encourage Farm Mechanization

- i) Population of the country escalating at 2.2% per year. Steps needs to be taken to fulfil the food demands of large population by **adopting intensive farming.**
Intensive farming requires machines on the farm
- ii) In multiple cropping system, **where high yielding variety of seeds are used,** all farm operations are required to be **completed in limited time with economy and efficiency.** This is possible with the help of mechanization.
- iii) **Farm mechanization removes drudgery of labour to a great extent.** A farmer has to walk about 66 km on foot while ploughing 1 ha land once by bullocks with a country plough having 15 cm furrow width
- iv) A **large number of females and children work on farm.** So, with mechanization females can work at home and children go to school.
- v) The **proper utilization of basic inputs like water, seeds and fertilizers** will be possible with proper equipment

- vi) There are certain **operations which are rather difficult to be performed by animal power or human labour such as:**
- a) Deep ploughing in case of deep-rooted crops.
 - b) Killing the pernicious weeds by deep tillage operations.
 - c) Levelling of uneven land.
 - d) Land reclamation.
- vii) **Application of insecticides, weedicide** etc

Benefits of farm mechanization

- **Timeliness** of operation
- Precision of operation
- Improvement of work environment
- Enhancement of **safety**
- Reduction of drudgery of labour
- **Reduction of loss of crops** and food products
- **Increased productivity** of land
- **Increased economic return** to farmers
- Improved dignity of farmers
- Progress and prosperity in rural areas

Limiting Factors in Farm Mechanization

- **Small and fragmented land holdings.**
- **Less investing** capacity of farmers.
- Agricultural **labour is easily available.**
- **Adequate draught animals** are available in the country.
- Lack of availability of suitable farm machines for different operations.
- Lack of **repair and servicing facilities** for machines.
- Lack of **trained man power.**
- Lack of **co-ordination between research organization** and manufacturers.
- High **cost of machines.**
- Inadequate quality control of machines

Bottlenecks In Indian Farm Mechanization System

- Low **annual use of tractors** (only **500-600 h/year** against recommended **1000 h/yr**).
- Non availability of matching equipment.
- Cumbersome and energy inefficient designs.
- **Poor reliability**, frequent breakdowns and high repair and maintenance cost.
- Use of **ungraded materials, absence of inter-changeability of components**.
- Inadequate **R&D, Testing & Training facilities** and inadequate Research funding.
- Inadequate user education.
- Lack of standardization.
- **Non-availability of relevant literature like operator's manual**, parts catalogues etc.

Priority Areas for Agricultural Mechanization in India

- **Intensification of R & D to introduce energy efficient machines** for relatively un mechanized crops such as cotton, sugarcane, oil seeds, pulses, vegetables & fruits. **Use reverse engineering and enforce close collaboration with farm machinery manufacturers.** Assist Indian manufacturers in seeking collaboration with well-known foreign firms wherever desired engineering technologies are not available.
- Intensify research in the **area of tractor design engineering due to their** extensive use in Indian farming. India is now the largest tractor manufacturer in the world. TMA needs to be involved in this task.
- An area of **utmost importance from environmental point of view is proper** utilization of about 540 million tonnes of crop residues available in India. Punjab alone has 10 million tonnes paddy straw which is mostly burnt. **Appropriate machines for incorporation of residues into the soil, for mulching, for collection, handling & transport for briquetting, gasification, power generation, and/or allied usage is a priority area in the field of mechanization.**
- Greater emphasis be **laid on design and manufacture of high capacity and precision machines for multi farm use, for corporate/contract farming** as well as for custom hiring through Agri. Business Centres being promoted by Govt. of India for the benefit of rural youths.
- **Mechanization of hill-agriculture (20% total cultivated area)**, horticulture and floriculture, forage production and handling equipment, forestry mechanization, and efficient transport equipment are some important areas.

- **Women-friendly tools and gadgets need to be evolved by modifying** the existing ones and designing the new tools to reduce drudgery to women workers.
- **Nearly two-third of the cultivated area is rainfed. Farm power available in these areas is barely 0.3 kW/ha.** Hence, mechanization of these areas should be under taken on priority basis. **Large horse power tractors and suitable equipment for conservation of soil moisture, seed bed preparation, seeding/planting, harvesting etc., are required.**
- There being a positive relationship between power availability and agricultural productivity, power constraint should be removed. **An annual growth rate of 4% over 1996 base in power supply to raise it from 1kW/ha to 2kW/ha by 2020 will be adequate to maintain a growth rate of 3% or more in agricultural production.** This is based on “**power production relation**” studies in India and abroad. The additional power will be supplied by tractors, power tillers, self-propelled machines, engines and electric motors.
- **Increasing emphasis on Integrated Pest management and Organic farming would require** use of efficient cultivation machinery for weeding and hoeing. Research in this area would be necessary to evolve optimum planting geometry and practices.
- **Under the WTO regime with liberalization of markets foreign** countries might take advantage of dumping their machinery in India, especially such equipment as sugar-cane harvesters, paddy transplanters, potato combines, cotton pickers, horticultural machinery, sprayers unless required equipment are expeditiously developed indigenously and have cost and quality competitiveness. **Joint projects by R&D organizations and Indian firms would be desirable.**

Cultural Operations in Crop Production:

About 70% of the Indian population practices agriculture. Hence, the production and management of crops is an important aspect to ensure optimal productivity in the fields.

The major agricultural operations involved in crop production and management are listed below:

1. Seedbed Preparation

The soil is **loosened and tilted before** the seeds are sown. Ploughs are used for the purpose. If the soil contains big lumps, **they are broken with the help of a hoe.** This

process aerates the soil so that the roots breathe easily. **The nutrients and minerals get properly mixed with the soil and come at the top.** Thus, the fertility of the soil increases and is fit for plantation.

2. Sowing/Planting/Transplanting

The **good quality, infection-free seeds are collected and sown on the prepared land.**

The seeds should be sown at proper depths and proper distances.

3. Manuring

The **soil may not have the right nutrients to efficiently sustain plant growth.** Hence, manures and fertilizers are added to the soil to increase its fertility and help plants grow better. **Manure is prepared by using decomposing plant and animal matter in compost pits.** Fertilizers, on the other hand, are **chemicals prepared in factories which contain nutrients for a specific plant.** They give faster results than manures. However, when excessively used, they turn the soil infertile

4. Irrigation

Crops require **water at regular intervals for proper growth.** The supply of water to the plants is known as irrigation. **Well, rivers, lakes, tube-wells are different sources for irrigation.** The traditional methods of agriculture involve the use of humans and animals. The **various traditional ways are moats, chain-pump, dhekli, rahat.**



Figure 7 – Seedbed Preparation



Figure 8 – Manuring



Figure 9 – Irrigation

The modern techniques of irrigation include the **sprinkler system and the drip system**.

5. Weeding & intercultural operation

The process of **removal of weeds is called weeding**. To achieve this, weedicides are employed, which are essentially chemicals specifically made to destroy weeds. **They are usually sprayed before seeding and flowering**.

Inter-culturing means culturing or tilling or stirring the soil in between the lines of a crop. This is done by an implement called **kolapi (a hoe)**. Inter-culturing is done to remove the weeds that **take away**



Figure 10 – Weeding

large quantities of moisture and plant food, to aerate the soil, to prepare a mulch (a loose layer of dry soil as covering), to conserve soil moisture useful for the crop, to prune the roots so as **to encourage a deep root system** and to kill the harmful insects hibernating in the soil.

6. Harvesting

When the crop matures, it is cut for further processing. **This process is known as harvesting**. It is usually manual labour, done **with the help of sickle**. However, mechanical harvesting is **used these days – machines such as combine harvesters** are used where the crops are harvested and threshed in one go.



Figure 11 – Harvesting

7. Threshing

The separation of grains from ear heads of the harvested crops is called as threshing. It is done either **mechanically or by cattle**.

Winnowing- The separation of grains and chaff is called winnowing. It is done either **mechanically or manually**.

Storage: The grains should be properly stored if they are to be kept for longer periods. They need to be protected from pests and moisture. The freshly harvested seeds should be dried before they are stored. **This prevents the attack from microorganisms and pests.**



Figure 12 – Threshing



Figure 13- Storage

Farm Machinery and Equipment

Farm machinery in agricultural field is used for land reclamation

Earth moving equipment **is heavy equipment, typically heavy-duty vehicles designed for construction operations which involve earthworks.** They are used to move large amounts of earth, to **dig foundations for landscaping** and so on. Earthmoving equipment **may also be referred to as; heavy trucks, heavy machines, construction equipment, engineering equipment, heavy vehicles and heavy hydraulics.** Most earthmoving equipment uses hydraulic drives as the primary source of motion. **Land reclamation, usually known as reclamation, and also known as land fill (not to be confused with a waste landfill), is the process of creating new land from oceans, seas, riverbeds or lake beds. The land reclaimed is known as reclamation ground or landfills.**

They are **essentially equipment to move the soil from one place to another** to modify the contour of field surface. Machinery used for developing the land to make it suitable for cultivation.

Earthmoving Operations Are Necessary For:

- Terrace building
- Filling of gullies or depressions
- Land smoothening for irrigation or surface drainage
- Construction of drainage ditches
- Building of earth dams
- Building of embankments
- Construction of ponds
- Irrigation channels

There Are Various Types of Earthmoving Equipment Used In Construction; Few Of The Pivotal Ones Are Described Below-

1. Backhoe dozer
2. Backhoe loader
3. Tracer blade
4. Excavator
5. Hydraulic scraper
6. Bulldozer
7. Laser guided land leveller

1. Backhoe Dozer

Used for **excavating soil, garbage handling, widening of rural roads** and removal of bushes and trees etc.

Agricultural land **levelling, making bunds & terracing of farm**, and site clearance

2. Backhoe Loader

A backhoe loader, also called a loader backhoe, digger in layman's terms, or colloquially shortened to backhoe within the industry, is a heavy equipment vehicle that consists of a tractor-like unit fitted with a loader-style shovel/bucket on the front and a backhoe on the back



Figure 14 – Blackhoe Dozer



Figure 15 – Blackhoe loader

3. Tracer Blade

It is attached to **the tractor with the 3-point linkage system** and is hydraulically controlled.

It consists of replaceable blade attached to curved steel body, side wings and **indexing arrangement for tilting and angling of the blade.**



Figure 16 – Tracer Blade

4. Excavators

Excavators are **heavy construction equipment** consisting of a **boom, dipper (or stick), bucket** and cab on a rotating platform known as the “house”

The **house sits atop an undercarriage with tracks or wheels.** They are a natural progression from the steam shovels and often mistakenly called power shovels

5. Bulldozer

A bulldozer is a tractor equipped with a **substantial metal plate (known as a blade)** used to push large quantities of soil, sand, rubble, or other such



Figure 17 - Bulldozer

material during construction or conversion work and **typically equipped at the rear with a claw-like device** (known as a ripper) to loosen densely compacted material.

6. Hydraulic Scraper

The **hydraulic scraper is towed behind the tractor**. The scraper consists of **cutting blade, hydraulic system, hitch point, hitch bar, apron, bowl, wheels, apron cylinder, and side frame**. It is used for collecting the soil from one place and unloading at the other. **It is used for rough levelling, cutting of high spots and filling of depressions.**



Figure 18 – Hydraulic Scraper

7. Laser Guided Land Leveler

Laser levelling **is a process of smoothing the land surface (± 2 cm)** from its average elevation using laser-equipped drag buckets.

This practice uses **large horsepower tractors and soil movers that are equipped with global positioning systems (GPS)** and/or laser-guided instrumentation so that the soil can be moved either by cutting or filling to create the desired slope/level.



Figure 19 – Laser Guided Land Leveler

This technique is well known for achieving higher levels of accuracy in land levelling and offers great potential for water savings and higher grain yields.

Tillage

Mechanical manipulation of soil to provide favourable condition for proper crop growth is called tillage.

Soil tillage **consists of breaking the compact surface of earth to a certain depth** and to loosen the soil mass so as to enable **the roots of the crops to penetrate** and spread into the soil.

Objectives of Tillage

To prepare a **desirable soil structure for a deep seed bed or a root bed suitable** for different types of crops.

- To **control weeds or to remove** unwanted crop plants (thinning).
- To **manage plant residues**. (Mixing of trash is desirable from the tilth and decomposition stand points and retention of trash in the top layers reduce erosion).
- To **minimize soil erosion**. (By following counter tillage, listing and proper placement of trash).
- To **establish specific surface configurations** for planting, irrigating, drainage, harvesting operations etc.
- To **incorporate and mix fertilizers**, pesticides, soil amendments etc. in to the soil.

Tillage Systems

Tillage system **consists of sequences of operations that manipulate the soil in order to produce a crop**. The operations include tilling, planting, fertilization, pesticide application, harvesting, and residue chopping or shredding. The ways in which these operations are implemented affect the physical and chemical properties of the soil, which in turn affect plant growth.

There are two types of tillage systems namely-

A) Conventional Tillage System

B) Conservation Tillage System

Conventional Tillage or Clean Tillage: Ploughing the entire field several times to prepare a seed bed is called conventional tillage

Conservation Tillage: Ploughing the **field with lesser number of passes** over the entire land or ploughing only in the **required space of the land and then sowing is called conservation tillage.**

Different types of conservation tillage are as follows

Minimum Tillage – Minimum soil manipulation necessary to meet tillage requirements.

Mulch Tillage – Tillage operations in which nearly 30 % of crop residue or other mulching materials are left on or near the soil surface is called mulch tillage.

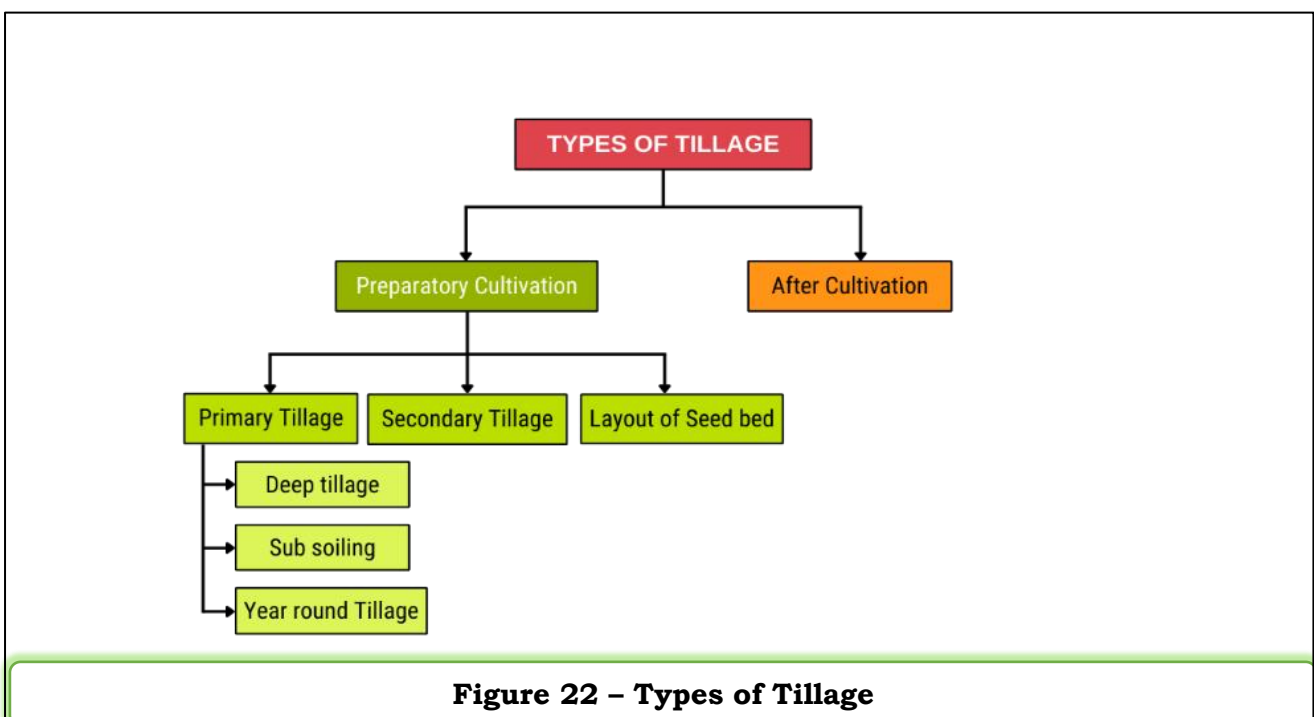
Rotary Tillage – Tillage operations employing rotary action of the tool to cut, break and mix the soil is called rotary tillage.

Strip Tillage - In strip tillage system only isolated bands of soil are tilled.

Combined Tillage – Tillage operations utilizing simultaneously two or more different types of tillage tools or implements to simplify, control or reduce the number of operations over a field is called combined tillage.

Classification Of Tillage

Tillage operations for seed bed preparations are classified as:



i) Preparatory Tillage

a. Primary Tillage –

- Deep Tillage
- Sub Soiling
- Year-Round Tillage

b. Secondary Tillage

ii) After Cultivation

Preparatory Tillage

This refers to tillage operations **that are done to prepare the field for raising crops**. It consists **of deep opening and loosening of the soil** to bring about a desirable tilth as well as to incorporate or uproot weeds and crop stubble when the soil is in a workable condition.

Preparatory Tillage Is of Two Types- Primary Tillage and Secondary Tillage



Figure 20 – Primary Tillage



Figure 21 – Secondary Tillage

After Cultivation

The tillage operations that are carried out in the standing crop are called after tillage. It includes drilling or side dressing of fertilisers, earthing up and inter-cultivation.

- **Earthing up** is an operation carried out with **country plough or ridge plough** so as to form ridges at the base of the crop. It is done either to **provide extra support against lodging as in sugarcane** or to **provide more soil volume** for better growth of tubers as in potato or to facilitate irrigation as in vegetables.
- **Inter-cultivation** is **working blade harrows, rotary hoes etc.**, in between the crop rows so as to control weeds. Inter-cultivation may also serve as moisture conservation measure by closing deep cracks in black soils.

Primary Tillage

The initial major operation designed to plough the soil deeply to reduce soil strength, cover plant materials and rearrange aggregates is called primary tillage.

The Objectives of Primary Tillage Are:

- a. To **reduce** soil strength
- b. To **rearrange aggregates**
- c. To **cover plant materials** and burry weeds
- d. To **kill insects and pests**

Types Of Primary Tillage

1. Deep Tillage

Deep ploughing **turns out large sized clods**, which are **baked by the hot sun** when it is done in summer. These clods crumble due to **alternate heating and cooling** and due to occasional summer showers. This process of **gradual disintegration of clods improves soil structure**. The rhizomes and tubers of perennial weeds (world's problematic weeds viz., *Cynodon dactylon* and *Cyperus rotundus*) die due to exposure to hot sun. Summer deep ploughing kills pests due to exposure of pupae to hot sun.

- **A deep tillage of 25-30 cm** depth is necessary for deep rooted crop like pigeon pea while moderate **deep tillage of 15-20 cm** is required for maize.
- **Deep tillage also improves soil moisture content.** However, the advantage of deep tillage in dry farming condition depends on rainfall pattern and crop.
- It is **advisable to go for deep ploughing only for long duration, deep rooted crops.** Depth of ploughing should be related to the amount of rainfall that it can wet.

The Implements Used for Primary Tillage Are Called as Primary Tillage Implements.

They include many **animals drawn** and **tractor drawn** implements.

- **Animal Drawn Implements include** -indigenous ploughs and mould-board ploughs.
- **Tractor Drawn Implements include**- mould-board ploughs, disc ploughs, subsoil ploughs, chisel ploughs and other similar implements.

Primary Tillage Equipment

1. Plough-

The main implement **used for primary tillage is a plough**. Ploughing essentially consists of opening the **upper crust of the soil, breaking the clods** and making the soil suitable for sowing seeds.

The purpose of ploughing can be summarized as follows

1. To obtain a **deep seed bed** of good texture.
2. To increase the **water holding capacity** of the soil.
3. To improve **soil aeration**.
4. To destroy **weeds and grasses**.
5. To destroy **insects and pests**.
6. To **prevent soil erosion**
7. To **add fertility to the soil** by covering vegetation.

Equipment's used for primary tillage – Indigenous Plough, Mouldboard plough, Disc plough, Reversible disc plough, Rotary tiller, Chisel plough, Sub-Soil Plough

1. Indigenous Plough-

It is an **animal-drawn plough**. It penetrates into the soil and breaks it open. It forms v shaped furrows with **15-20 cm top width and 12-15 cm** depth. It can be used for ploughing in the dry land, garden land and wetlands. In villages, local artisans make the plough and supply it to the farmers. **These ploughs are also called country ploughs**



Figure 23 – Country Plough

2. Mouldboard Plough

Mouldboard plough is one of the oldest of all agricultural implements and is generally considered to be the important tillage implement. **Ploughing accounts for more traction energy than any other field operation.**

Mouldboard ploughs are available for animals, power tiller and tractor operation.



Figure 24 – Mould-Board Plough

While working, a mouldboard plough does four jobs namely

- a) **Cutting** the furrow slice
- b) **Lifting** the furrow slice
- c) **Inverting** the furrow slice and
- d) **Pulverizing** the furrow slice.

3. Disc Plough

An action of a disc plough is similar to the mould board plough. Disc plough **cuts, turns and in some cases breaks furrow slices** by means of separately mounted large steel concave discs. A disc plough is designed with a view **to reduce friction by making a rolling plough bottom instead of sliding plough bottom as** in the case of mould board plough. A disc plough works well in the conditions where mould board plough does not work satisfactorily.

Advantages Of Disc Plough

- A disc plough **can be forced to penetrate into the soil which** is too hard and dry for working with a mouldboard plough.
- It **works well in sticky soils** in which a mouldboard plough does not scour.

- It is more useful for deep ploughing.
- It can be **used safely in rough, stony and stumpy soils** without much danger of breakage.
- A **disc plough works well even after a considerable** part of the disc is worn out in abrasive soils.
- It works well in **loose soils also (such as peat)** without much clogging.



Figure 25 – Disc Plough

Disadvantages Of Disc Plough

- It is **not suitable for covering surface trash** and weeds as effectively as a mould board plough
- Comparatively, **a disc plough leaves the soil in rough and cloddy condition** than that of a mould board plough.
- Disc **plough is much heavier than mould board plough** for equal capacities because penetration of disc plough is affected largely by its weight rather than suction. (**Mould board plough is forced into the soil by the suction of the plough, while the disc plough is forced into the soil by its own weight**).

4. Reversible Disc Plough

- It is similar to **standard disc plough**, except that it can turn over the furrow slice to left or right side.
- It **not only improves the rate of work but also leaves the field flat and level**.
- This plough is **found very successful for contour ploughing** because the **operation** can be performed back and forth on the contour line without any problem.



Figure 26 – Reversible Disc Plough

- The soil is turned in **the direction required to prevent soil erosion**. Slopes can also be gradually levelled just by ploughing down the slope every year.
- It **saves water and time, improves irrigation efficiency and ensures** an even crop growth and no water wastage. The discs are rotated from one side to another by using a lever.

5. Rotary Tiller

The rotary cultivator **is widely considered to be the most important tool** as it **provides fine degree of pulverization** enabling the necessary rapid and intimate mixing of soil **besides reduction in traction demanded by the tractor driving wheels** due to the ability of the soil working blades to provide some forward thrust to the cultivating outfit



Figure 27 - Rotary Tiller

6. Chisel Plough

Chisel ploughs are used to break through and shatter compacted or otherwise impermeable soil layers. **Deep tillage shatters compacted sub soil layers and aids in better infiltration and storage of rainwater in the crop root zone.** The improved soil structure also results in better development of root system and the yield of crops and their drought tolerance is also improved.



Figure 28- Chisel Plough

7. Sub-Soil Plough

The function of the **sub-soiler** is to **penetrate deeper** than the conventional cultivation machinery **and break up the layers of the soil**, which have become compacted due to the movement of heavy machinery or as a result of continuous ploughing at a constant depth. These **compacted areas prevent the natural drainage of the soil** and also inhibit the passage of air and nutrients through the soil structure.



Figure 29 – Sub-Soil Plough

Secondary Tillage

Lighter and finer tillage operations performed in the soil after primary tillage to create proper soil tilth and surface configuration for seeding and planting are called secondary tillage operations.

Secondary tillage operations are generally done on the surface soil. **They do not cause much soil inversion and shifting of soil from one place to other.** They consume less power per unit area compared to primary tillage operations. **These operations consume less power per unit area compared to primary tillage operations.**

Secondary tillage implements may be tractor drawn or bullock drawn implements.

The Main Objectives of Secondary Tillage Are

The implements used for secondary tillage operations are called secondary tillage implements. **They include different types of harrows, cultivators, sweeps, clod crushers, levellers, bund formers, ridge ploughs etc.**

1. Harrow

Harrow is a secondary tillage implement that **cuts the soil to a shallow depth** for **smoothing and pulverizing the soil as well** as to cut the weeds and to mix the materials with the soil.

There are several types of harrows used in India such as **disc harrow, spike tooth harrow, spring tooth harrow, acme harrow, patela, triangular harrow, bade harrow, guntaka and reciprocating power harrow.**

Disc Harrow- Disc harrow is found very suitable for hard ground, full of stalks and grasses. It cuts the lumps of soil, clods and roots.

Spring Tooth Harrow-It is a harrow with **tough flexible teeth, suitable to work in hard and stony soils.** This type pulverizes the soil and helps in killing weeds.

Acme harrow- The front part of the knife breaks the soil and crushes the clods. This **harrow obtains a good pulverization.** It is good for mulching also.



Figure 30 – Disc Harrow

Patela-It is a wooden plank used for **smoothing the soil and crushing the weeds.** It is also used for breaking clods, packing and levelling the ploughed soil and to remove the weeds.

Blade harrow (Bakhar)-It is used to prepare **seed beds mostly in clayey soils.** It works like a sweep, which moves into top surface of the soil without inverting the soil.

Guntaka-It is **used for shallow working of the soil** with minimum soil inversion. It is mainly used to prepare the seed bed mostly in clay soils.

Reciprocating power harrow-It has **two horizontal oscillating arms** having staggered pegs in two rows at 200mm spacing.

2. Cultivators

It is an **implement for inter cultivation with laterally adjustable tines or discs to work between crop rows.** The cultivator stirs the soil, and breaks the clods. The tines fitted on the frame of the cultivator comb the soil deeply in the field. A cultivator performs

functions intermediate between those of plough and the harrow. **Destruction of weeds is the primary function of a cultivator.**

The following are a few important functions performed by a cultivator. Inter culture the fields.

- **Destroy the weeds** in the field.
- Aerate the soil for **proper growth of crops.**
- **Conserve moisture by** preparing mulch on the surface.
- To **sow seeds when it is provided with sowing** attachments.
- To prevent **surface evaporation and encourage** rapid infiltration of rain water into the soil.



Figure 31 - Cultivator

3. Bund Former

It is used for **making bunds or ridges by collecting the soil.** Bunds are required to hold water in the soil, **thereby one can conserve moisture and prevent run-off.** The size of the bund former is determined by measuring the maximum horizontal distance between the two rear ends of the farming boards. Bund former consists of forming board, beam and handle.



Figure 32 – Bund Former

4. Ridger

It is an implement **importantly used to form ridges required for sowing row crop seeds and plants in well-tilled soil.**

The ridger is also used for forming field or channels, earthing up and similar other operations. **Ridger is also known as ridging plough and double mould board plough.** The **ridger generally has 'V' shaped or wedge-shaped share** fitted to the frog. The nose or tip of share penetrates into the soil and breaks the earth. The mould boards lift, invert and **also cast aside the soil, forming deep channels and ridges** of the required size.



Figure 33 – Ridger

5. Puddler

Puddler is an **implement for churning the soil with water.** It is used to **prepare paddy fields with standing water** after initial ploughing with country plough. It breaks up the clods and churns the soil. **The main purpose of puddling is to reduce leaching of water**

or decrease percolation of water, to kill the weeds by decomposition and to facilitate transplantation of paddy seedlings by making the soil softer.

- Puddling is done in standing **water of 5-10 cm** depth.
- A common puddler has **puddling units each having four straight blades** or paddles or fan type blades or squirrel type blade mounted on an axle.



Figure 34 - Puddler

6. Leveller

Land levelling is **expected to bring permanent improvement in the value of land**. Levelling work is **carried out to modify the existing contours of land** so as to achieve certain objectives desired for efficient agricultural production system.

These Objectives Include

- **Efficient application** of irrigation water
- **Improved surface** drainage
- **Minimum soil erosion**
- Increased** conservation of rain water specially on dry lands
- Provision of an **adequate field size and even topography** for efficient mechanisation



Figure 35 – Land Leveller

Sowing, Planting & Transplanting Equipment

Sowing- is an art of placing seeds in the soil to have good germination in the field.

Perfect sowing gives

- Correct number of seed per unit area.
- Correct depth of sowing
- Correct spacing between row-to-row and plant to plant.
- Correct seed rate

Sowing Methods

- (i) Broadcasting
- (ii) Dibbling
- (iii) Drilling
- (iv) Seed dropping behind the plough
- (v) Transplanting
- (vi) Hill dropping
- (vii) Check row planting.

1. Broadcasting

Broadcasting is the **method of random scattering of seeds on the surface of seedbed.**
It can be done manually or mechanically.

- When broadcasting is done **manually, uniformity of seed placement depends upon the skill of the man scattering the seeds. Soon after broadcasting the seeds are covered by planking or some other devices.**
- **Usually, higher seed rate** is obtained in this system.



Figure 36 – Broadcasting Of Seeds

- **Mechanical broadcasters are used for large-scale sowing.** The device scatters the seeds on the surface of the seedbed at controlled rates.

2. Dibbling

Dibbling is **the process of placing seeds in holes made in the seedbed and closing the seed with soil.** In this method, seeds are placed in holes made at definite depth at fixed spacing.

- The equipment used for **dibbling is called dibbler.**
- It is a conical shape instrument used to make proper holes in the field.
- Small hand dibblers are made with several conical projections made in a frame.
- This is **very time-consuming process**, so it is not suitable for small seeds. Mostly vegetables are sown in this way.



Figure 37 – Dibbling Of Seed Using Dibbler

3. Seed Dropping Behind the Plough

It is a very common method of sowing followed by farmers in villages.

- **This method is used for seeds like maize, gram, peas, wheat and barley.**
- A woman/ man walks behind a plough ploughing the land and drop the seeds in the furrows made by the plough. **Sowing behind the plough can be done by a device known as malobansa.**
- It consists of a bamboo tube provided with a funnel shaped mouth.



Figure 38 – Seed Dropping Behind Plough

- **It is fitted to the handle of the plough.** One man drops the seeds through the funnel and other man handles the plough and the bullocks. This method is a slow and laborious method.

4. Drilling

Drilling consists of dropping the seeds in furrow lines in a continuous stream and covering them with soil.

The spacing between the **seeds is not uniform**. Seed metering may be done either manually or mechanically.

The number of rows planted may be one or more

This method **is very helpful in achieving proper depth of sowing, proper spacing between seeds and proper seed rate**. Drilling can be done by **using seed drills of tractor drawn and animal drawn types**.



Figure 39 – Seed Driller

5. Transplanting

Transplanting consists of rising the seedlings in a nursery bed and then planting the seedlings in another field (main field).

- It is commonly done **for paddy, vegetable and flowers**.
- It is a **time-consuming** operation.
- **Equipment used** for planting the seedlings in the main field is **called transplanter**.

6. Hill Dropping

In this method, few seeds are dropped as a **hill at a fixed place and not in a continuous stream**. **The spacing between hill to hill in a row is constant**. The equipment's are called planters.



Figure 40 – Hill Dropping of Seed

7. Check Row Planting

It is a method of planting, **in which row-to-row and plant-to-plant distance is uniform.** In this method, seeds are planted **precisely along straight parallel furrows.** The rows are always in two perpendicular directions. **A machine used for check row planting is called check row planter.**

Seed Drill

Seed drill is a machine used for placing the seeds in a continuous stream in furrows at uniform rate and at controlled depth with an arrangement of covering the seeds with soil.

According to the power source used, seed drills may be classified into

- (i) Bullock drawn seed drills
- (ii) Tractor drawn seed drills.

According to the type of seed metering done animal drawn seed drills may be classified into

1. **Manually metered seed drills-** In manually metered seed drills a person drops the seeds in the furrows
2. **Mechanically metered seed drill-** In mechanically metered seed drills a mechanical device called seed metering mechanism is used to meter the seeds.
Functions of a seed drill:

Seed Drill Performs the Following Functions

- i) To carry the seeds.
- ii) To open furrows at uniform depths
- iii) To meter the seeds
- iv) To drop the seeds in furrows in an acceptable pattern
- v) To cover the seeds and compact the soil around the seed.

Equipment Used

1. Seed Cum Fertilizer Drill

Seed drills fitted with fertilizer dropping attachments are called seed-cum-fertilizer drills, they deliver both the seeds and fertilizers simultaneously in an acceptable pattern. Seed cum fertilizer drill has a large seed box which is divided lengthwise into two compartments, one for seed and another for fertilizers distribution.



Figure 41 – Seed Cum Fertilizer Drill

Functions Of A Seed Cum Fertilizer Drill

Seed cum fertilizer drill performs the following functions

1. To carry the **seeds and fertilizer in separate** compartments.
2. To open **furrows at uniform depths**
3. To **meter the seeds** and fertilizers
4. To deposit the **seed and fertilizer in the furrows** in an acceptable pattern
5. To **cover the seed and fertilizer** and compact the soil around the seed.

2. Planter

Planter is sowing equipment used for sowing those seeds which are larger in size and cannot be handled by seed drills. Row to row and plant to plant spacing is maintained in a planter. **Potato planter, maize planter and cotton planters are popularly used**

Functions Of a Planter:

- i) To **open the furrow**

- ii) To **meter the seed**
- iii) To **deposit the seed in the furrow**
- iv) To cover **the seed and compact** the soil over the seed.

2. Transplanter

Transplanter is **the equipment used for transplanting mat-type paddy seedlings in the main field**. It is suitable for **all types of paddy varieties grown by transplanting**. Both manually operated and mechanically operated transplants are available. Before transplanting, **the field has to be puddled well and allowed** for one or two days for consolidation of soil so that the machine can work without sink age.



Figure 42 – Paddy Transplanter

3. Zero-Till Seed-Cum-Fertilizer Drill

No-till farming (also called zero tillage or direct drilling) is an agricultural technique way of growing crops or pasture without disturbing the soil through tillage. Proponents **claim no-tillage increases the amount of water that infiltrates into the soil, the soil's retention of organic matter, and its cycling of nutrients**. In a few agricultural regions, it may reduce soil erosion.



Figure 43 – Zero Till Seed Cum Ferti Drill

It increases the amount and variety of life in and on the soil, including disease-causing organisms and disease organisms.

4. Zero-Till Drill-cum-Bed Planter

Zero-till drill-cum-bed planter is a prototype which has been developed by combining a **zero-till seed-cum-fertilizer drill** with a **bed planter**. Efforts have been made to incorporate the functions of **these two separate machines** in one combined unit. It can be used to **harness the benefits of both zero-till seed-cum-fertilizer drill** as well as of a bed planter and guidelines and instructions for its use should be followed accordingly.

5. Strip Till Drill

Strip-till is a conservation system that uses a **minimum tillage**. It combines the soil drying and warming benefits of conventional tillage with the soil-protecting advantages of no-till by **disturbing only the portion of the soil that** is to contain the seed row. This type of tillage is performed with special equipment and can require the farmer to make multiple trips, depending on the strip-till implement used, and field conditions. Each row that has been strip-tilled is usually about eight to ten inches wide



Figure 44 – Strip Till Drill

6. Vegetable Transplanters

Transplanters are designed based on **seedling type to be used**. The semiautomatic transplanters can be used for **almost all types of seedlings**. The bare root seedlings are obtained by pulling seedlings directly from nursery beds.

The soil block seedlings are obtained by **sowing of seeds in moist soil cubes made by mixing soil, peat, compost, and sand**.



Figure 45 – Potato Planter

Tractor

Tractor is a self-propelled power unit having wheels or tracks for operating agricultural implements and machines including trailers. Tractor engine is used as a prime mover for active tools and stationary farm machinery through power-take off (pto) or belt pulley.

Classification Of Tractors:

Tractors can be classified into three classes on the basis of structural design:

1. Wheel tractor
2. Crawler tractor
3. Walking type tractor



Figure 46- Wheel Tractor



Figure 47- Crawler Tractor



Figure 48- Walking Type Tractor

Wheel tractor: Tractors having **three or four pneumatic wheels** are called wheel tractors. Four-wheel tractors are popular everywhere.

Crawler tractor: This type is also called Track type tractor or Chain type tractor. In such tractors, **there is endless chain or track in place of pneumatic wheels.**

Power tiller: Power tiller is a walking type tractor. This tractor is usually fitted with

two wheels only. The direction of **travel and its controls for field operation is performed by the operator, walking behind the tractor.**

Selection Of Tractor

Selection of tractors depend up on following factors

Land holding: Under a single cropping pattern, it is normally recommended to consider 1 hp for every 2 hectares of land. In other words, one tractor 20-25 hp is suitable for 40-hectare farm.

Cropping pattern: Generally, 1.5 hectare/hp has been recommended where adequate irrigation facility are available and more than one crop is taken. So, a 30-35 hp tractor is suitable for 40 hectares of land.

Soil condition: A tractor with less wheel base, **higher ground clearance** and low overall weight may work successfully in lighter soils **but will not be able to give sufficient depth in black cotton soils.**

Climatic Condition: For **very hot zone and desert area, air cooled engines** are preferred over water cooled engines. Similarly for higher altitude air cooled engines are preferred because water cooled engines are liable to be frozen at high altitudes.

Repair facilities: It should be ensured that the tractor to be purchased has a dealer at nearby place with all the technical skills **for repair and maintenance of the machine.**

Running cost: Tractors with **less specific fuel consumption** should be preferred over others so that the running cost may be less.

Initial cost and resale value: While **keeping the resale value in mind**, the initial cost should not be very high, otherwise higher amount of interest have to be paid.